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SHORT COMMUNICATION/COURTE COMMUNICATION

Clinical and mycological evaluation of an herbal antifungal formulation in canine *Malassezia* dermatitis



Traitement de la dermatite à Malassezia chez le chien au moyen d'huiles essentielles : étude clinique et mycologique

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pachydermatis;
Essential oils;
In vitro susceptibility;
In vivo treatment

Summary

Objective. — *Malassezia pachydermatis* is a common cause of more widespread dermatitis in dogs (CMD). Recurrences are common, and this disorder can be very troubling for both dogs and for the pet owner.

Material and methods. — The treatment of 20 dogs affected by dermatitis due to *M. pachydermatis*, with Malacalm[®], a commercially available mixture consisting of essential oils (*Citrus aurantium* 1%, *Lavandula officinalis* 1%, *Origanum vulgare* 0.5%, *Origanum majorana* 0.5%, *Mentha piperita* 0.5% and *Helichrysum italicum* var. *italicum* 0.5%, in sweet almond oil and coconut oil) is reported. The effectiveness of the whole mixture, of component essential oils and of their more represented compounds against clinical isolates was evaluated by a microdilution test. Twenty animals were topically administered the mixture twice daily for 1 month. Ten animals were treated with a conventional therapy based on ketoconazole 10 mg/kg/day and chlorhexidine 2% twice a week for 3 weeks. At the end of both treatments animals significantly improved their clinical status. Adverse effects were never noticed. Follow-up visit performed on day 180th allowed to observe a recurrence of clinical signs in all the subjects treated conventionally, while not significant clinical changes were referred in dogs treated with Malacalm[®]. The overall MIC value of Malacalm[®] was 0.3%. *O. vulgare* showed the lowest minimum inhibitory concentrations (MIC), being active at 0.8%, followed by *M. piperita* (1%), *O. majorana* (1.3%),

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MOTS CLÉS

Malassezia pachydermatis ;
Huiles essentielles ;
Sensibilité in vitro ;
Traitement in vivo

C. aurantium (2%) and *L. officinalis* (4%) while *H. italicum* did not yield any antimycotic effect up to 10%. Active major compounds were thymol, carvacrol, p-cymene, 1,8-cineol, limonene and menthol.

Conclusion. — The phytotherapeutic treatment achieved a good clinical outcome, and no recurrence of skin disorders on day 180th was recorded. This herbal remedium appeared to be a safe tool for limiting recurrences of CMD.

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Résumé

Objectif. — Hôte habituel de la flore cutanée chez le chien, *Malassezia pachydermatis* est une levure qui est capable dans certains cas, de se multiplier de manière anarchique et de provoquer ainsi une dermatite. Beaucoup d'options thérapeutiques sont envisageables dans le traitement de cette infection mais cela nécessite temps et argent. Et de plus, les récurrences sont fréquentes. **Matériels et méthodes.** — L'objectif de l'étude est d'évaluer l'efficacité clinique de cette mycose vis-à-vis d'un mélange d'huiles essentielles composé de *Citrus aurantium* 1 %, *Lavandula officinalis* 1 %, *Origanum vulgare* 0,5 %, *Origanum majorana* 0,5 %, *Mentha piperita* 0,5 % et d'*Helichrysum italicum* var. *italicum* 0,5 %, en huiles d'amande douce et de noix de coco comme adjuvant. Le produit est disponible dans le commerce sous le nom de Malacalm® et il s'applique en topique deux fois par jour durant 1 mois lorsqu'une dermatite à *Malassezia* est diagnostiquée chez un chien. La sensibilité in vitro de 5 souches de *M. pachydermatis* a été évaluée vis-à-vis de la mixture des composants et des substances les plus représentées. En parallèle, 10 chiens ont été traités par le kétoconazole (10 mg/kg, une fois par jour durant 21 jours) en association avec une solution de chlorhexidine à 2 % (deux fois par semaine pendant 21 jours).

Résultats. — Au terme des deux protocoles, les animaux traités présentent une amélioration clinique de leur état et aucun effet secondaire n'est constaté. Après six mois, le groupe d'animaux traités avec l'application de kétoconazole et de chlorhexidine présente une rechute. La CMI du Malacalm® est de 0,3 %. L'huile essentielle d'*O. vulgare* se révèle la plus active (CMI 0,8 %), suivie par *M. piperita* (1 %), *O. majorana* (1,3 %), *C. aurantium* (2 %) et *L. officinalis* (4 %). L'activité antifongique d'*H. italicum* n'est efficace qu'à partir d'une concentration à 10 %. Les composants les plus actifs sont le thymol, le carvacrole, le p-cymène, le limonène et le menthol. **Conclusion.** — Le traitement de la dermatite à *Malassezia* chez le chien à l'aide d'huiles essentielles procure une amélioration clinique et il ne présente aucun effet secondaire après six mois d'application. L'application de ce mélange peut être employée afin de limiter les rechutes de cette infection mycosique.

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Introduction

Malassezia pachydermatis is a lipophilic yeast, which is part of the cutaneous microflora of several warm-blooded vertebrates. It commonly occurs on the skin and mucosae of healthy dogs and the factors involved in the transition from commensalism to parasitism are not fully understood, even if dogs with skin disease associated with *M. pachydermatis* may have 100–10.000-fold increases in skin population densities, compared to healthy dogs [2]. Yeasts overgrowth could be promoted by excessive sebum production and/or disruption of epidermal barrier such as maybe occurring in hypersensitivity diseases, keratinization disorders, ectoparasites infections, bacterial pyoderma and endocrine disequilibrium [3]. Descriptions of *M. pachydermatis* as a common cause of more widespread dermatitis in dogs are now fully accepted by veterinary dermatologists [2,8].

Current therapeutic options for canine *M. pachydermatis* dermatitis (CMD) relay on topical shampoo consisting of 2% miconazole nitrate and 2% chlorhexidine [10,13]. When the use of these drugs is contraindicated or impractical in a clinical setting, systemic oral ketoconazole, itraconazole [7], fluconazole, and terbinafine [1] should be employed.

Recurrent cases of CMD are common. The identification and correction, when possible, of predisposing factors is often essential for successful management, although many dogs with *Malassezia* dermatitis require regular maintenance therapy to prevent relapse. For all these reasons, this disorder can be very troubling for dogs and time-consuming, frustrating an extremely costly for the pet owner.

In recent years, the interest in selecting sustainable products from landscape plants has increased and some data are available for *Malassezia* species of both human and veterinary interest indicating extracts of several plants as active in vitro against *Malassezia* spp. [6,12,18]. However, there are no data about the use of essential oils (EOs) in the treatment of dermatitis due to *M. pachydermatis* in dogs.

The aims of the present paper were:

- to assess the in vitro antifungal activity of a commercially available mixture of EOs (Malacalm®, Flora s.r.l., Pisa, Italy), of its chemically defined EOs components and of the more represented compounds against *M. pachydermatis*;
- to evaluate the effectiveness of the treatment under field conditions by a prospective, randomized, single blinded clinical trial.

Materials and methods

Malacalm[®] is a commercially available mixture composed by the following EOs: *Citrus aurantium* 1%, *Lavandula officinalis* 1%, *Origanum vulgare* 0.5%, *Origanum majorana* 0.5%, *Mentha piperita* 0.5% and *Helichrysum italicum* var. *italicum* 0.5%, in sweet almond oil and Coconut oil, proposed as adjuvant treatment in dermatological disorders. In the present trial this product was administered as sole therapy to treat CMD.

In vitro assay

The in vitro anti-*M. pachydermatis* activity of Malacalm[®] as a whole product was assayed by a microdilution test, conversely each single component and the compounds present in higher amounts were also evaluated. Malacalm[®] and all employed EOs were supplied by Flora s.r.l (Lorenzana, Pisa, Italy).

Volatile constituents of each EO were analyzed by GC-MS as previously reported [15]: Briefly, a CP-3800 gas chromatograph equipped with HP-5 capillary column (30 m × 0.25 mm; coating thickness, 0.25 mm) and Varian Saturn 2000 ion trap mass detector were employed. Analytical conditions were as follows: injector and transfer line temperature, 220 and 240 °C respectively; oven temperature, programmed from 60 to 240 °C at 3 °C/min; carrier gas, helium at 1 ml/min; injection, 0.2 ml (10% hexane solution); split ratio, 1:30. The constituents were identified by comparing the retention time with the authentic samples, on the basis of their linear indices relative to a series of *n*-alkanes (C8-C23). Further identifications were carried out using a homemade library of mass spectra created from pure substances and components of known oils, and the MS literature data (NIST 2000, ADAMS).

EOs antifungal activity was tested against five clinical isolates of *M. pachydermatis*, obtained from recurrent cases of CMD. For all in vitro studies Malacalm[®] as well as EOs from *C. aurantium* s.130187, *H. italicum* s. 1101204, *L. officinalis* s.128660, *M. piperita* s.80059, *O. vulgare* s.60326, *O. majorana* s. 80717, sweet almond oil (*Prunus dulcis* Mill) s.70312 and coconut oil (*Cocos nucifera*) s.763.600.39, were assayed. Tests were performed as reported elsewhere [15], with slight modification. Briefly, all stock solutions were prepared in sweet almond and coconut oil and diluted to obtain concentrations ranging from 0.1 to 10%. The dilutions were chosen considering that EOs in vivo maximum dilution for dermatologic administration is 20%. Control cultures were achieved with medium alone, and with medium supplemented with diluent oils. Fungal growth was daily evaluated until complete development was observed in control wells after 7 days. To avoid bias due to differing amounts of sweet almond and coconut oils, each test was carried out by using a concentration equal to 40% of them. These concentrations were also added to the medium in control cultures. After incubation at 30 °C, yeasts inocula were harvested by centrifugation, gently washed in fresh mDixon broth, then inoculated on mDixon agar plates to monitor killing of the yeasts. Malacalm[®] was assayed to calculate the overall MIC value, also. Furthermore, to evaluate the efficacy of a conventional antimycotic drug, recommended for the treatment of CMD, the yeasts were tested against ketoconazole by E-test (AB Biodisk, Solna, Sweden) [14].

In order to evaluate the antimycotic activity of components present in higher amounts and/or known for their antifungal activity, clinical isolates were incubated with p-cymene, 1,8-cineole, α -e- β -pinene, γ -terpinene, thymol, carvacrol, menthol, menthone, menthofuran, trans-sabinene hydrate, limonene and linalool diluted in sweet almond and coconut oil (0.01–10%), in semisolid Dixon agar, as previously reported for EOs. All tests were carried out in triplicate. All products were provided by Sigma-Aldrich (Milan, Italy). The viability of the inocula was evaluated as described above.

In vivo treatment

Thirty-five privately owned dogs, affected by CMD, of various breeds and crossbreeds and both genders, aged between 1 and 12 years (median 4 years) were included in the study, after obtaining informed owner's consent form. All dogs have been previously administered conventional anti-*Malassezia* treatments, with consistent recurrence within 3 to 4 months after cessation of treatment. Criteria for inclusion were the presence of local or generalized dermatitis, associated with cytological detection of greater than two *Malassezia* organisms/high power fields (400×) [7]. Animals treated from less than 4 months before the inclusion day (day 0), and/or with concomitant dermatoses were not admitted to the study. Skin lesions were evaluated and scored as reported by Maynard et al. [10]. The animals were randomly divided into 3 groups. Group A (20 dogs) was administered Malacalm[®]. Dogs were treated twice a day directly on affected areas for 1 month. Group B consisted of 10 animals treated with a conventional therapy based on ketoconazole 10 mg/kg/day and chlorhexidine 2% twice a week for 3 weeks. Animals from group C were 5 untreated controls.

Malassezia population size was determined by means of a semiquantitative swab technique [3] from affected body areas. Six areas were examined for frequency of yeasts occurrence, more in detail axillae (75%, 50% and 60% for group A, B and C, respectively), groin (90%, 90% and 100%), interdigital areas/nail beds of forelimbs (95%, 100% and 100% and 90%, 90% and 100% respectively), and interdigital areas/nail beds of hind limbs (85%, 80% and 100% and 90%, 100% and 100%, respectively). Fungal load was determined by microscopic count in 10 fields (400×)/area with values up to 64 yeast cells, with overall prevalence of 30.2% (group A), 30.7% (group B) and 29.9% (group C). Cultures were performed onto mDixon medium and dermatophyte selective medium for species identification, as previously reported [15].

Follow-up visits were performed on days 30 and 180 for all groups, by clinical and cytological evaluation, as well as by culture and by clinical evaluation only, respectively. The owners were advised to refer any adverse effect. Clinical outcome was considered good in presence of more than 50% of reduction of aggregate score of clinical signs, mild to moderate with a reduction from 10% to 50%. When reduction was less than 10% the treatment was referred as ineffective.

Statistical analysis was performed by Wilcoxon rank sum test to evaluate differences in yeast count and clinical score, before and after treatments.

Results

In vitro assay

The chemical composition of the tested oils is reported in Table 1. The main components were typical monoterpenes such as thymol, γ -terpinene and p-cymene (*O. vulgare*, *O. majorana*), carvacrol (*O. vulgare*), 4-terpineol and trans-sabinene hydrate (*O. majorana*), limonene and β pinene (*C. aurantium*), menthol, menthofuran and menthone (*M. piperita*), linalool (*L. officinalis* and *C. aurantium*) while in *H. italicum* the major constituent was α -pinene.

O. vulgare showed the lowest minimum inhibitory concentrations (MIC), being active at 0.8%, followed by *M. piperita* (1%), *O. majorana* (1.3%), *C. aurantium* (2%) and *L. officinalis* (4%) while *H. italicum* did not yield any antimycotic effect up to 10%.

The overall MIC value of Malacalm[®] was 0.3%. The MIC for ketoconazole was $< 0.03 \mu\text{g}/\text{ml}^{-1}$ for all yeasts isolates. MIC values corresponded to MFCs for all tested EOs.

Active major compounds were thymol, carvacrol, p-cymene, 1,8-cineol, limonene and menthol. Complete data about the MICs/MFCs are reported in Table 2.

In vivo treatment

On day 30th treated animals of both groups A and B significantly improved their clinical status. Adverse effects were never noticed. Eighteen animals (90%) from group A showed a good outcome, the other 2 animals had a moderate (25%) resolution of clinical signs. Conversely, all the subjects from the group B had a good clinical improvement. Dogs from group C did not show any significant variation in their clinical status. Significant differences were not observed between the 2 treatment groups.

Semiquantitative microscopical evaluation showed a marked reduction of frequency of yeasts occurrence in 90% of axillae (both groups), in 95% of groins for group A and 100% for group B. Conversely only 30% of interdigital areas and 50% of nail beds of forelimbs (both groups), in 55% and 40% of interdigital areas (group A and B, respectively) and in 45% and 30% of nail beds of hind limbs (group A and B, respectively) resulted microscopically negative. Large reductions of *Malassezia* counts were observed in all animals of treated groups (overall prevalence 2.6% and 2.9%, respectively), except for 3 animals from group A and 1 from group B. Two among these animals, belonging to group A, had shown a moderate clinical resolution. All high *Malassezia* population sizes after treatment were observed in interdigital areas and nail beds of both forelimbs and hind limbs. Untreated controls did not show any significant change in yeast population size.

Malassezia colonies were cultured from all microscopically positive animals on day 0 and on day 30th. The species *M. pachydermatis* was identified in all dogs and at all anatomic sites.

Clinical and cytological outcomes were significantly different when comparing values at days 0 and 30th, respectively in group A and group B ($P \leq 0.01$). Follow-up visit performed on day 180th allowed to observe a recurrence of clinical signs in all the subjects from group B, while not

significant clinical changes were referred in dogs belonging to group A.

Discussion

To our knowledge this is the first in vitro evaluation of *M. pachydermatis* sensitivity to some EOs, corroborated by a clinical trial to treat CMD. All active EOs were at the same concentration both fungistatic and fungicidal, except for *H. italicum* that did not show any antifungal effect. However its anti-inflammatory and antioxidant properties [17] make it an interesting component in dermatological mixtures. In the assays of sensitivity to antifungal drugs used for conventional treatment MIC values can depend on a number of factors [16], furthermore the discrepancies among results obtained by different authors could be ascribed also to a lack of standard reference techniques for in vitro testing, due to fastidious growth requirements of this yeast in lab conditions.

Active major compounds appeared to be carvacrol, menthol, thymol, p-cymene and limonene as reported in literature [4,19], while 10% β -pinene, γ -terpinene, menthofuran and trans-sabinene hydrate did not exert any antifungal activity. Surprisingly, the MIC value for Malacalm[®] as whole product was very low (0.3%) when compared to MICs of each EOs components, ranging from 0.8% to up to 10%. This finding could be explained by some synergistic effects among different EOs. Attention should be paid when applying in vitro results to clinical practice, considering that in CMD underlying dermatological disorders can make skin more sensitive. Although our results cannot be compared with other similar, the lack of adverse effects would suggest a potential safety of this product.

Both phytotherapeutic and conventional treatments achieved a good clinical and cytological outcome, even if 2 dogs from group A did not undergo to a complete clinical cure. However, the good health status did not last for conventionally treated animals on day 180th, who all showed recurrence of skin disorders. It should be noticed that all the enrolled animals had previously been treated with conventional anti-*Malassezia* drugs, relapsing when the treatment was discontinued. Recurrences are in fact one of the most troublesome aspects of CMD. If recurrences are frequent, shampoos and/or topical rinses on a once- or twice-weekly basis or oral ketoconazole every 3 days could be employed [11], but this treatment requires a good pet owner's compliance and the use of ketoconazole for long term therapy can provoke adverse effects [9]. The herbal remedy was well-accepted from both pets and owners, who moreover appreciated good smell of Malacalm[®], efficiently fighting the strong rancid odor characteristic of CMD.

Clinical recovery was usually associated with a decrease in *Malassezia* population size, except for 2 animals from group A and B respectively, who showed a good clinical improvement with a reduction of yeasts counts in axillae and groins only. In 2 dogs belonging to group A, a partial reduction of amounts of *Malassezia* was associated with a moderate clinical response.

Antifungal activity of Malacalm[®] seems to be enhanced in respect to its single components. EOs in various combination can show both synergistic and antagonistic effects and their

Table 1 Chemical composition of the analyzed essential oils.
Composition chimique des huiles essentielles analysées.

Compound	LRI ^a	<i>C. am.</i>	<i>H. it.</i>	<i>L. of.</i>	<i>M. pi.</i>	<i>O. vu.</i>	<i>O. ma.</i>
Heptanal	902		0.1				
α -Thujene	930	0.1	0.1	0.1		1.4	1.0
α -Pinene	939	1.3	30.9	0.6	1.0	0.8	0.9
Camphene	954	0.1	0.9	0.5		0.2	0.1
Thuja-2,4(10)-diene	960		0.1				
Sabinene	975	1.5	tr	0.2	0.7		3.7
β -Pinene	979	13.6	0.8	0.8	1.6		0.8
3-Octanone	984			tr		0.1	0.1
6-Methyl-5-hepten-2-ol	989	0.1	tr				
Myrcene	991	1.2	0.2	0.5	0.2	2.2	1.6
1-Octanol	992			0.2	0.3		
α -Terpinene	1003		0.3			2.3	
α -Phellandrene	1007					0.3	0.2
δ -3-Carene	1013					0.1	
p-Cymene	1025	0.6	0.6	0.4	0.5	14.3	7.6
Limonene	1029	17.7	3.8	0.8	2.1	0.4	2.9
β -Phellandrene	1031					0.3	tr
1,8-Cineole	1033			7.7	6.8	0.4	tr
(Z)- β -Ocimene	1037	0.1	tr	0.2	0.2	0.2	0.1
(E)- β -Ocimene	1050	0.7	0.1	0.1	tr	0.4	tr
γ -Terpinene	1060		0.6		0.4	11.2	6.4
cis-Sabinene hydrate	1070	0.1		0.1	1.1	0.2	3.2
cis-Linalool oxide furanoid	1089	1.1		0.3			
Terpinolene	1089		0.3		0.2	0.3	1.6
2-Nonanone	1090		0.1				
trans-Sabinene hydrate	1097						12.1
trans-Linalool oxide furanoid	1098	0.9		0.4			
Linalool	1099	37.5	1.3	31.1	0.6	3.2	
cis-Thujone	1102		0.5				0.1
Octadienal<2,2-dimethyl-3,4>	1104			0.5			
Fenchol	1120		0.1				
cis-Limonene oxide	1137	0.2					
cis-p-Menth-2,8-dien-1-ol	1138			tr			0.7
trans-Pinocarveol	1139	0.4	0.1	0.1			
Camphor	1146		0.2	7.0		0.3	0.3
Menthone	1053				23.6		
Neryl oxide	1159		0.1				
Menthofuran	1164				11.8		
Borneol	1169		0.3	6.5		0.8	0.3
4,6-Dimethyloctane-3,5-dione	1171		0.1				
Menthol	1172				33.9		
cis-Linalool oxide pyranoid	1176	0.1					
trans-Linalool oxide pyranoid	1177	0.1					
4-Terpineol	1177	0.6	0.5	3.9	0.3	1.8	16.0
m-Cymen-8-ol	1183	0.1		0.1		0.2	0.1
α -Terpineol	1189	4.9	0.5	1.2	0.3	0.7	2.7
Nerol	1190		0.4				
neo-iso-Verbanol	1191						0.2
cis-Dihydrocarvone	1193		0.1				
Myrtenal	1196	0.2					
trans-Carveol	1198	0.1					
Verbanol	1198						0.3
1-Octyl acetate	1202			0.1			

Table 1 (Continued)

Compound	LRI ^a	<i>C. am.</i>	<i>H. it.</i>	<i>L. of.</i>	<i>M. pi.</i>	<i>O. vu.</i>	<i>O. ma.</i>
Nerol	1230	0.8		0.2			
Pulegone	1237				2.3		
Carvone	1243						0.7
Geraniol	1253	5.2					
Piperitone	1253				0.5		
Linalyl acetate	1257			22.2			3.3
Geranial	1267			0.1			0.2
Bornyl acetate	1286		0.6				
Lavandulyl acetate	1290			2.8			
Thymol	1290					45.0	20.2
2-Undecanone	1294		0.1				
Menthyl acetate	1295				5.1		
Carvacrol	1299					4.0	
δ-Elemene	1338	0.1					
α-Terpinyl acetate	1349	0.2					
Eugenol	1359						0.1
Neryl acetate	1362	1.7	4.8	0.3		0.1	
α-Copaene	1377		2.9			tr	0.1
Geranyl acetate	1383			0.6		0.3	
β-Bourbonene	1388				0.4	0.2	0.1
β-Elemene	1391	3.0					
Italicene	1406		3.4				
α-Gurjunene	1410			0.1			
cis-α-Bergamotene	1413		1.0	0.1			
β-Caryophyllene	1419	0.3	4.4	1.3	2.3	1.3	1.6
trans-α-Bergamotene	1435		0.9	0.2		tr	
4,6,9-Trimethyldec-8-ene-3,5-dione	1444		2.1				
α-Humulene	1455	0.1	0.3	0.1		0.1	0.4
Neryl prpanoate	1455		0.4				
ε-β-Farnesene	1457			0.8	0.2	0.2	
Alloaromadendrene	1460		0.3			0.2	
α-Acoradiene	1466		0.6				
β-Acoradiene	1471		0.1				
γ-Murolene	1480		1.6			0.2	0.1
γ-Curcumene	1483		9.0				
Germacrene D	1485			0.1	1.8	tr	0.6
β-Selinene	1496		8.7				
α-Selinene	1498		3.2				
Bicyclogermacrene	1500				0.4		0.2
α-Murolene	1500		0.3				
trans-Nerolidol	1500	1.9					
β-Bisabolene	1502			0.2		2.6	1.4
δ-Amorphene	1512		0.1				
γ-Cadinene	1514			0.7		0.3	0.1
β-Curcumene	1520		1.2				
δ-Cadinene	1523		0.7	0.1			0.3
β-Sesquiphellandrene	1525					tr	
Italicene ether	1531		0.2				
α-Cadinene	1539		tr			tr	
Caryophyllene oxide	1583	0.3		0.4		0.1	0.5
Neryl isovalerate	1584		0.4				
Spathulenol	1593	0.1				0.1	0.4
Viridiflorol	1593				0.4		
2,4,6,9-Tetramethyldec-8-ene-3,5-dione	1598		0.1				
5-Eudesmen-11-ol	1606		0.1				
1,10-di- <i>epi</i> -Cubenol	1619			0.1			
Humulene oxide II	1621		0.1				
γ-Eudesmol	1632		tr				

Table 1 (Continued)

Compound	LRI ^a	<i>C. am.</i>	<i>H. it.</i>	<i>L. of.</i>	<i>M. pi.</i>	<i>O. vu.</i>	<i>O. ma.</i>
β-Eudesmol	1651		tr				
α-Eudesmol	1654		0.1				
β-Bisabolol	1675		tr				
<i>Epi</i> -α-Bisabolol	1685			0.5			
Total		97.0	89.8	94.3	99.0	96.8	93.3

C. am.: *Citrus aurantium*; *H. it.*: *Helichrysum italicum*; *L. of.*: *Lavandula officinalis*; *M. pi.*: *Mentha piperita*; *O. vu.*: *Origanum vulgare*; *O. ma.*: *Origanum majorana*; tr: traces.

^a Linear Retention Index calculated on a DB-5 ms column.

Table 2 MIC values of compounds more represented in Malacalm[®] versus *Malassezia pachydermatis*. Valeurs des CMI des substances les plus représentées du Malacalm[®] vis-à-vis de *Malassezia pachydermatis*.

Compound	MIC (%)
Thymol	0,025
Carvacrol	0,05
p-cymene	1
1,8-cineole	1
Limonene	1
Menthol	1
Linalool	2,5
Menthone	5
α-pinene	5
β-pinene	> 10
γ-terpinene	> 10
Trans-sabinene hydrate	> 10
Menthofuran	> 10

MIC: minimum inhibitory concentration.

use as mixture is a recent approach. The effects of the same mixture can differ on the basis of the chemical composition of the single components and the tested organism [5], consequently the chemical characterization of constituents is strictly required before performing a clinical trial.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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